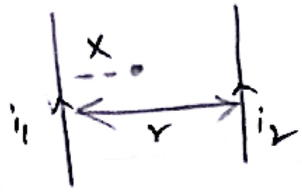
LESSON-3

Moving Charges and Magnetism
Electromagnetic Induction and AC

1. Null Point:



$$x = \frac{r}{\left(\frac{i_2}{i_1}+1\right)}$$
(Same direction)

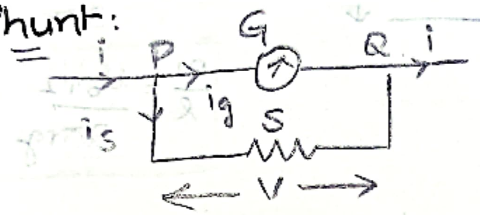
$$x = \frac{r}{\left(\frac{i_2}{i_1}-1\right)}$$
(Opposite direction)

2. $\bar{\tau} = \bar{M}\times\bar{B}$ $\quad \bar{F} = q(\bar{V}\times\bar{B})$

$\tau = BiNA\sin\theta$. ($\theta \rightarrow$ Angle b/w Normal & $\bar{B}$)

$= BiNA\cos\alpha$. ($\alpha \rightarrow$ Angle b/w Plane of Coil & $\bar{B}$)

3. Shunt:



$R_{eq} = \frac{GS}{G+S}$ $\quad V = iR_{eq} = i\frac{GS}{G+S}$

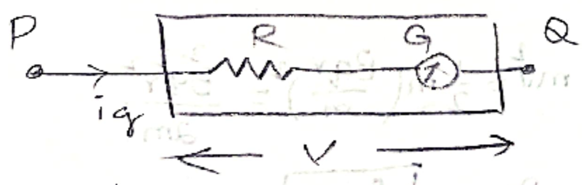
$V_{PQ} = i_g G = i_s S$

$S = \frac{G}{n-1}$ where $n \rightarrow n = \frac{i}{i_g} = \frac{\text{new range}}{\text{old range}}$

$i = i_g + i_s$

$i_g = \frac{iS}{G+S}$ $\quad i_s = \frac{iG}{G+S}$, $\boxed{\frac{i_g}{i_s} = \frac{S}{G}}$

4. Voltmeter:



$V = i_g(G+R)$

$R = \frac{V}{i_g} - G$.

$R = G(n-1)$ (Connect in Series)

$C\theta = BiAN \Rightarrow i = \left(\frac{C\theta}{BAN}\right)$ [$\because \theta \rightarrow$ Deflection for Passage of i]

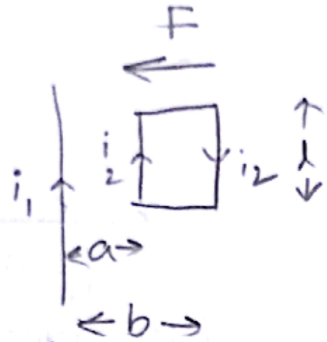
"C" $\rightarrow$ Couple per unit twist.

5. Work done :

$$W = MB(\cos\theta_1 - \cos\theta_2).$$

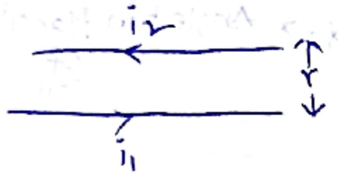
6. Forces:

1.



$$F = \frac{\mu_0 i_1 i_2 l}{2\pi}\left[\frac{1}{a} - \frac{1}{b}\right].$$

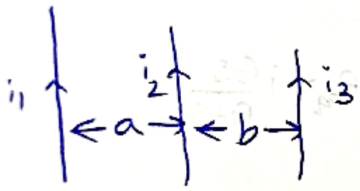
2.



$$F = mg \Rightarrow \frac{\mu_0 i_1 i_2 l}{2\pi r} = mg$$

$$\boxed{\frac{m}{l} = \frac{\mu_0 i_1 i_2}{2\pi r g}}$$

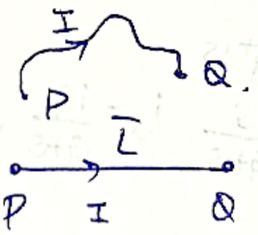
3.



$$F = \frac{\mu_0}{2\pi}\left[\frac{i_1 i_2}{a} - \frac{i_2 i_3}{b}\right].$$

4. $F_{res} = i\bar{L} \times \bar{B}$

$$= i\left(\int d\bar{l}\right) \times \bar{B}$$



7. Cyclotron:

- $KE = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{Bqr}{m}\right)^2 = \frac{B^2q^2r^2}{2m}.$
- $T = \frac{2\pi r}{v} = \frac{2\pi m}{Bq}.$ $\boxed{f = \frac{1}{T}}$
- $\frac{mv^2}{r} = Bqv$
- $\boxed{r = \frac{mv}{Bq}} = \frac{\sqrt{2mK}}{qB} = \frac{\sqrt{2mqV}}{qB} = \frac{mv\sin\theta}{qB}.$
- $\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mKE}} = \frac{h}{\sqrt{2 \ldots}}$

Special Cases:

1. Magnetic Induction $\bar{B}$:

Semi circular wire : $\frac{\mu_0 i}{4R}$.

$$B = \frac{\mu_0 i}{4\pi d}(\sin\alpha + \sin\beta).$$

$$\phi = \bar{B}\cdot\bar{A} = BA\cos\theta.$$

$$\phi = \int \bar{B}\cdot ds.$$

$$\phi = NBA\cos\theta.$$

$$M_{Max} = \sqrt{L_1 L_2}$$

$$\boxed{M = K\sqrt{L_1 L_2} \text{ Henry}}.$$

$$i = i_0(1 - e^{-t/\lambda}).$$

$$\omega = \frac{1}{\sqrt{LC}}, \quad f = \frac{1}{2\pi\sqrt{LC}}.$$

- $\boxed{\frac{E_1}{E_2} = \frac{N_s}{N_2} = \frac{I_2}{I_1}}$ $\left(\begin{matrix} 1 \rightarrow \text{Primary} \\ 2 \rightarrow \text{Second} \end{matrix}\right)$.

$$M_R = \sqrt{M_1^2 + M_2^2 + 2M_1M_2\cos\theta}. \quad \boxed{T = 2\pi\sqrt{\frac{I}{MB}}}.$$

Magnetic Moment $\boxed{|M| = NiA}$

$\boxed{\text{Magnetic Torque: } \bar{T} = \bar{M}\times\bar{B}}$

$\boxed{F = VBL}$

Terminal Velocity Net force $= 0$.

$\boxed{F_M = BiL}$

Proton:	q	m
Deutron:	q	2m
α Particle:	2q	4m.

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